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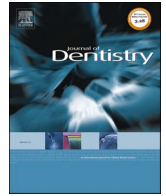
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Survival of molar teeth in need of complex endodontic treatment: Influence of the endodontic treatment and quality of the restoration

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1. Introduction

The goal of root canal therapy is to clean the root canals and reduce the bacterial load, in order to create an asymptomatic and stable tooth complex. Due to the complex anatomy of the root canal system, which can hinder complete cleaning, inflammation can persist or reoccur after some years and retreatment might be necessary. Endodontic retreatment is more complex and negatively influences the endodontic healing, often

necessitating referral to a specialist setting [1].

Besides endodontic treatment the quality of the final restoration plays an important role in endodontic healing and tooth survival [2,3]. Several restorative factors improve the prognosis for tooth survival of endodontically treated teeth, such as a cusp coverage restoration, the presence of mesial and distal contacts and type of tooth [3]. The odds for tooth retention of endodontically treated non-molar teeth was 1.26 (95 % CI: 1.01–1.41) times greater than for molar teeth. In case of

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endodontically treated teeth, the loss of hard tissue due to fracture or caries is often the main cause for extraction, ranging from 46.4%–63.8% of the cases [4,5].

Besides tooth retention, the endodontically treated tooth should be free of inflammation. Endodontic healing can be evaluated by a combination of clinical and radiographic examinations. According to the Glossary of Endodontic Terms by the American Association of Endodontists, endodontic treatment outcome can be assessed as healed, healing or nonhealed [6]. When there are no clinical symptoms such as pain, palpation sensitivity, localized pockets, sinus tracts or mobility and radiographically there is no periapical pathosis, the case is considered 'healed'. Radiographically, endodontic healing can be assessed using the Periapical Index (PAI) [7]. It may take several years before a periapical radiolucency has resolved completely. In a consensus report of the European Society of Endodontology, a period of 4 years after endodontic treatment is advised to assess post treatment disease [8]. Therefore, a reduction in the size of a periapical radiolucency is a sign of endodontic 'healing'. When there are clinical symptoms or when there is no difference or an increase in size of the periapical radiolucency, a case is considered 'nonhealed' [9].

Several variables influence the endodontic healing process. Apart from the type of root canal treatment (primary or secondary treatment), the prognosis for endodontic healing is further influenced by six factors [10]: 1) the presence of a preoperative radiolucency, 2) presence of complicating operative factors (e.g. broken instruments or perforations), 3) quality of the coronal restoration, 4) distance of the root canal filling to the radiological apex of the tooth, 5) preservation of the root canal morphology and 6) compactness of the root canal filling. The last three factors can be summarized as the 'technical quality' after endodontic treatment. A root canal therapy of good technical quality means that the canal filling was compact, ended within 0–2 millimeters of the apex and the morphology was preserved.

In order to assess the difficulty of the endodontic treatment, guidelines were developed for the general practitioner: the Endodontic Treatment Classification score (ETC) [11]. This guideline aids the general practitioner when to refer a patient to an endodontic specialist. The ETC scale divides the level of difficulty of the endodontic treatment into three classes. Class I indicates an endodontic treatment of average risk, with limited preoperative concerns. A molar tooth with 3 canals without calcified canals or root curvatures and easy to isolate with rubber dam is an example of an ETC class I score. Class II represents a high risk of complications during the endodontic procedure and achieving a predictable treatment outcome will be difficult for an experienced clinician. A molar tooth with >3 canals, moderate curvature of the roots (10°–30°) and calcified canals is considered ETC class II. When scored as a class III, there is a very high risk of procedural complications and advanced knowledge and specific instruments are needed to achieve a predictable treatment outcome. A molar tooth in need of retreatment and the presence of a broken instrument is an example of a ETC class III.

The primary objective of this retrospective practice-based study was to evaluate the survival of molar teeth with ETC scores of II and III within a referral setting with a minimum service time of the root canal treatment of 12 months. Secondary objectives were to evaluate the quality of the coronal seal, the restoration and the endodontic treatment. The null hypothesis was that the type of endodontic treatment, the ETC score and the quality of the coronal seal would have no influence on tooth survival. Secondary hypothesis was that there would be no difference in restoration quality between the direct and indirect restorations.

2. Material and methods

Endodontically treated maxillary and mandibular first and second molars treated between January 2011 and October 2017 were included in this study. In this period, a total of 416 root canal treatments were performed on endodontically treated molars by two dentists in a referral

setting limited to endodontics. Inclusion criteria were molar teeth with a primary or secondary root canal treatment referred to and treated at the dental department of the Martini Hospital Groningen, patients older than 18 years of age and with an American Society of Anaesthesiologists (ASA) score of ≤ 2 . M teeth with open apices, combined non-surgical and surgical treatment, an ETC score of 1 or patients younger than 18 years of age or with an ASA-score >2 were excluded (Fig. 1).

2.1. Endodontic procedure

All endodontic procedures were performed under the microscope (OPMI Pico; Zeiss, Oberkochen, Germany) and rubber dam isolation (Isodam heavy; Sigma Dental, Handewitt, Germany) by two experienced dentists in endodontic (re)treatment in a referral clinic. After endodontic opening, working length was determined using an electronic apex locator (Root ZX II; Morita, Osaka, Japan). Root canals were shaped up to working length using hand files (ISO 6–20; Ready Steel K-file; Dentsply Maillefer, Tulsa, Oklahoma, USA) and a rotary file system (Protaper; Dentsply Sirona, York, Pennsylvania, USA) according to the instructions of the manufacturer. In endodontic retreatment cases, rotary files D1/D2/D3 (Protaper system, Dentsply Sirona) were used. Root canals were irrigated using a 2% sodium hypochlorite solution (Reymerink, 's-Gravelande, the Netherlands) and 17 % EDTA (Calasept EDTA, Directa, Väsby, Sweden). As a final irrigation protocol, sodium hypochlorite was activated using ultrasonic passive irrigation during 20 s, which was repeated for three times. After drying the canals using paper points, a root sealer (AH26 Root canal sealer, Dentsply Sirona) was applied to the canal walls. An apical gutta-percha (Protaper system, Dentsply Sirona) plug was created using master cone which was sealed off within 5 mm of working length (with the Continuous Wave of Obturation Technique). The remaining part of the root canal was filled using warm gutta-percha (BeeFill 2 in.; VDW GmbH, München, Germany) with a back-fill technique. A post-endodontic build-up was created using a sixth generation bonding agent (Clearfil SE Bond; Kuraray Noritake, Okayama, Japan) a bulk-fill composite (Surefil SDR flow; Dentsply Sirona, Milford, USA) to seal the coronal canal entrance and a nanohybrid composite (Filtek Supreme XTE; 3 M ESPE, Seefeld, Germany) via the incremental layering technique. After the endodontic treatment, an intraoral radiograph was taken to evaluate the post-operative endodontic quality.

2.2. Recall appointment

As a part of regular care, 9–12 months after endodontic intervention a recall appointment was planned to assess the healing process. During this recall appointment, patients were asked about possible symptoms. Endodontic diagnostics, both clinically (percussion, palpation, probing) and radiologically, were carried out. In case of symptoms, patients were rescheduled for an appointment after 12 months. After the recall appointment, patients returned to their general dentist for regular care. Because of the referral setting, after the recall appointment no further follow-up was present. In order to assess tooth survival, the healing after a period of 9–12 months and the quality of the restoration, an additional appointment was necessary. In a systematic review of tooth survival for all types of endodontically treated teeth, the cumulative survival rate was 86.7 % after 8–10 years. [3] In order to estimate a survival proportion of 86.7 % endodontically treated molars with 95 % confidence and a margin of error of 3.5 %, 362 endodontically treated molars needed to be included. This study was approved by the Medical Ethical Committee of the Martini hospital and registered in the national trial register (NL7153). Informed consent was obtained from all patients attending the clinic. Patient, pre-, post-endodontic and restorative variables were listed (see Table 1). Four observers (A.R., C.F., J.B. and M. K.), were calibrated for the endodontic evaluations. Light pictures were taken from occlusal, lingual and buccal side of the endodontically treated tooth using a Digital Single Lens Reflex camera (Canon 80D, 100

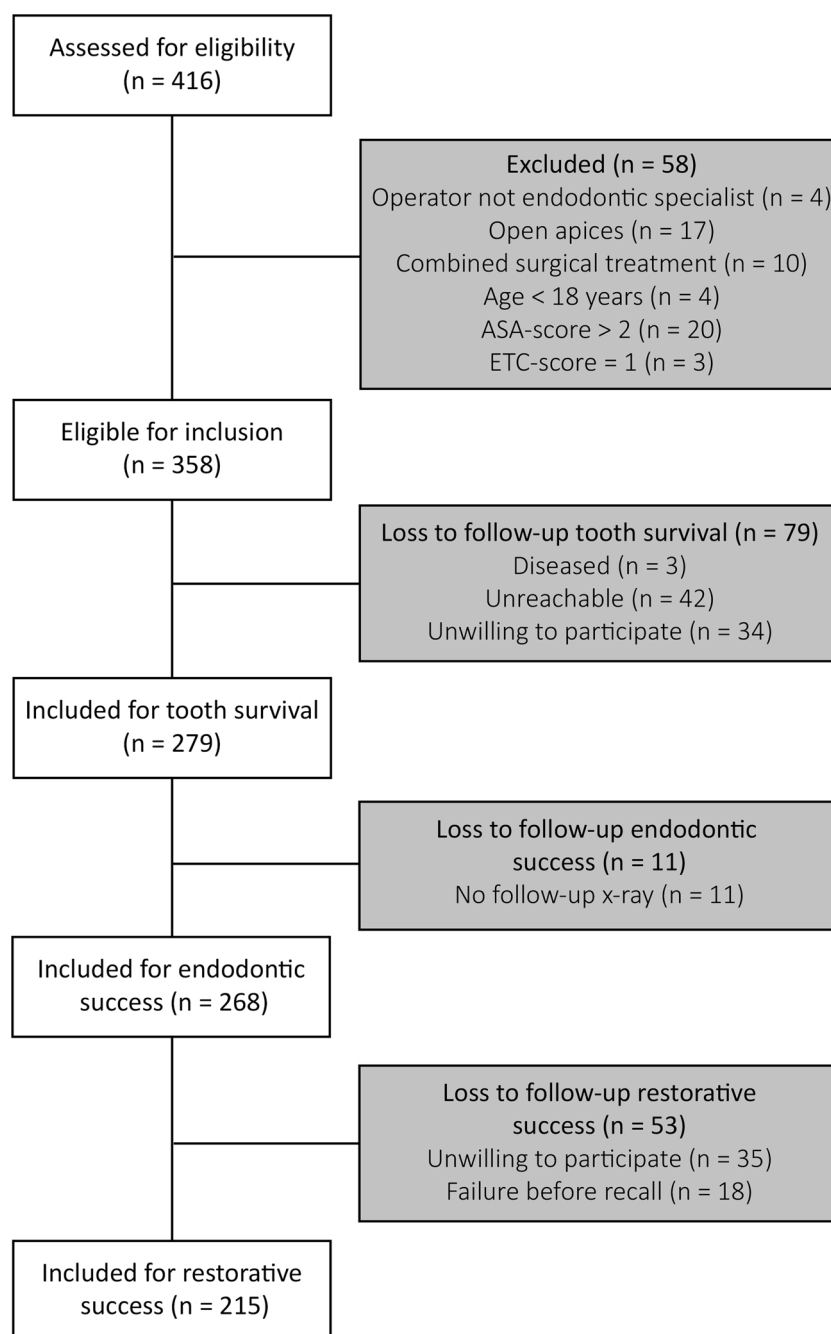


Fig. 1. Flow chat of inclusion process of endodontically treated molar teeth.

mm F2.8 macro; Canon Incorporation, Tokyo, Japan). When the radiograph was older than 2 years, the patient experienced clinical symptoms, there was a radiolucency present on the latest radiograph or when the coronal seal was compromised, a new radiograph was taken. Two observers were present during the appointment, one of whom was the same (M.K.) for all clinical observations. A joined decision was taken concerning findings during the clinical examination.

When a patient was unable to attend, an anamnesis was taken over the phone and patients' permission was asked to contact the general practitioner about the clinical and radiographic status of the tooth. For endodontic healing, the date of the latest radiograph in the chart of the general practitioner or the registration at the hospital was noted.

2.3. Patient variables

Patient variables included age, caries risk and parafunction. When a patient did not experience any caries during the two years before the recall appointment, the patient was considered as a low caries risk patient. When a patient had one caries lesion, the patient was considered as medium risk and when two or more lesion developed during the last two years, patients were scored as high caries risk.

2.4. Endodontic variables

Pre-endodontic variables consisted of the DETI/ETC score (I/II/III), endodontic diagnosis (irreversible pulpitis / apical periodontitis), number of root canals treated, preoperative complications (broken instrument, perforation, obliteration), preoperative radiolucency and the

Table 1

Variables included in the study. ETC: Endodontic Treatment Classification.

*Total number of endodontically treated molars n = 268.

Factor	Level	Number (%)
Operator	1	238 (85 %)
	2	41 (15 %)
Caries risk	Low	155 (55 %)
	Medium	78 (28 %)
	High	44 (16 %)
	Unknown	2 (1%)
Parafunction	Yes	85 (31 %)
	No	190 (68 %)
	Unknown	4 (1%)
Gender	Male	125 (45 %)
	Female	154 (55 %)
Jaw	Upper	145 (52 %)
	Lower	134 (48 %)
Number of canals treated	3	42 (15 %)
	4	237 (85 %)
ETC score	II	49 (18 %)
	III	230 (82 %)
Preoperative radiolucency	Yes	237 (85 %)
	No	42 (15 %)
Preoperative complicating factor	Yes	125 (45 %)
	No	154 (55 %)
Type of endodontic treatment	First	128 (46 %)
	Retreatment	151 (54 %)
Technical quality endodontic treatment	Adequate	179 (64 %)
	Inadequate	100 (36 %)
Restoration type follow-up	Direct	127 (46 %)
	Indirect	147 (52 %)
	Unknown	5 (2%)
Cusp coverage follow-up	Yes	179 (64 %)
	No	87 (31 %)
	Unknown	13 (5%)
Coronal seal follow-up	Adequate	230 (82 %)
	Inadequate	36 (13 %)
	Unknown	14 (4%)
Endodontic healing*	Healed	175 (65 %)
	Healing	49 (19 %)
	Nonhealed	44 (16 %)

type of endodontic treatment (initial/retreatment). Post-endodontic variables were the presence of clinical symptoms (anamnesis, percussion, palpation testing and probing), healing (healed/healing/non-healed), postoperative radiolucency and the technical quality of the endodontic treatment as scored on a radiograph. Healing was assessed using the endodontic treatment outcomes as was defined the Glossary of Endodontic Terms by the American Association of Endodontists. A case was considered 'healed' when no clinical symptoms and no periapical radiolucency was present. A case was considered 'healing' when no clinical symptoms were present and the peri-apical radiograph showed a reduction in size of the radiolucency as compared to baseline. When clinical symptoms were present or the radiolucency did not reduce or increased in size or intervention was necessary (extraction, non-surgical or surgical (re)treatment), a case was considered as 'nonhealed'. Post-operative radiolucency was assessed using the Periapical Index (PAI). [7] A PAI ≥ 3 was scored as a postoperative radiolucency. When the obturation of the root canals was compact (no voids), followed the original morphology and finished within 0–2 mm of the radiographic apex, the technical quality was scored as adequate. When one of the criteria was inadequate, the overall technical quality was judged to be inadequate.

2.5. Restorative variables

Restorative variables were the jaw, presence of adjacent teeth, cusp coverage of all remaining cusps, the quality of the coronal seal and restoration quality as scored by the FDI criteria [12], restoration surfaces, type (direct/indirect) and material (composite/amalgam/glass ceramic/PFM/zirconia). The FDI criteria grades the restoration on

esthetic, functional and biological aspects. When scored in the range of 'clinically excellent' to 'clinically sufficient', there is no need for the restoration to be replaced. A score of 'clinically unsatisfactory' suggest replacement of the restoration for prophylactic reasons and a score of 'clinically poor' indicates a necessary replacement. Observers were calibrated for the FDI criteria using the online calibration tool www.e-calib.info.

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